CQI Monitoring System

Use Case Diagram — Program !' PEO / PO !' Course / LO + Visualization

ROLE-BASED STARTER • React + Vite + Supabase (RLS)

NBSC · Institute of Computer Studies •
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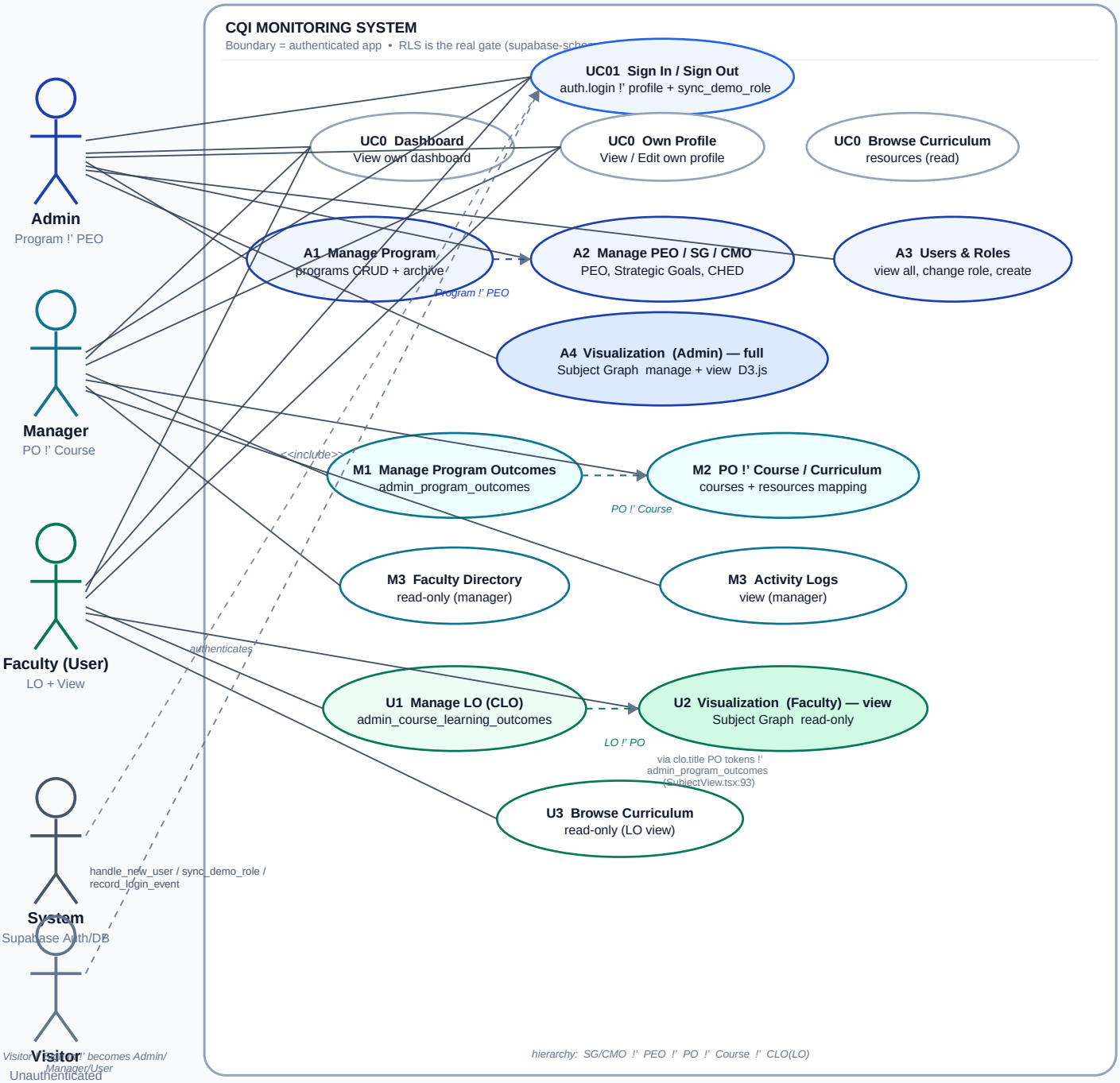
docs/use-case-diagram.pdf • A4 portrait • vector



Role Matrix & Mapping Notes

How Program!'PEO / PO!'Course / LO+Visualization maps to tables, RLS, and pages

Source: README.md Role Matrix • supabase-schema.sql RLS • src/App.tsx NAV

Capability	Admin	Manager	Faculty (User)	Evidence
Sign in / out, Dashboard, Own Profile	yes	yes	yes	Login.tsx, App.tsx
Browse Curriculum (resources read)	yes	yes	yes	resources SELECT = authenticated
Manage Program	yes CRUD+D	—	—	programs / Program.tsx
Manage PEO / SG / CMO	yes CRUD+D	—	—	admin/*, RLS admin only
Manage Program Outcomes (PO)	yes	yes C/U*	—	admin_program_outcomes
Manage Course / Curriculum	yes	yes C/U*	—	courses + resources
PO !' Course mapping	—	yes	—	po links via peo_id/sg_id/cmo_id
Manage LO (CLO) - create/edit	yes	yes	yes	admin_course_learning_outcomes
Manage LO - delete	yes	—	—	DELETE = admin only
Visualization (Subject Graph)	full (manage+view)	—	view-only	SubjectView.tsx D3
Faculty Directory	yes	yes (read)	—	manager/Users.tsx
Activity Logs	yes	yes (view)	—	activity_logs SELECT admin/mgr
Users & Roles (change role/create)	yes	—	—	admin/Users.tsx + admin-users fn

* C/U = create/update only; delete = admin only (supabase-schema.sql:381,390,413). D = delete.

Outcome Hierarchy — how the visualization connects (SubjectView.tsx:5)

SG (strategic_goals) !' CMO (ched_memo_orders) !' PEO (program_educational_objectives) !' PO (admin_program_outcomes) !' Course (courses + curriculum) !' CLO/LO (admin_course_learning_outcomes)

Subject Graph: Subject (double-ring) !' Curriculum + Prereq/Coreq !' CLOs !' POs !' PEO/SG/CMO (SubjectView.tsx:20)

Responsibility Slice (your requirement)

Admin — Program !' PEO

Owns programs, PEOs, SG, CMO, users/roles, and full visualization (manage+view).

Manager — PO !' Course

Owns Program Outcomes and Course/Curriculum mapping; PO!'Course via peo_id/sg_id/cmo_id links.

Faculty (User) — LO + View

Owns LO (CLO create/edit) and read-only visualization + curriculum browse.

Notes — RLS & gating (read the lines, not just the boxes)

- RLS is the real gate — hiding buttons is UX only. Policies in supabase-schema.sql use current_user_role().
- resources: SELECT = authenticated, INSERT/UPDATE = admin|manager, DELETE = admin.
- admin_course_learning_outcomes: INSERT/UPDATE = admin|manager|user, DELETE = admin (LO slice).
- activity_logs: SELECT = admin|manager, INSERT via log_activity() SECURITY DEFINER (stamps JWT email).
- Gating in src/App.tsx NAV + PAGES map coerces activePage to role-valid page.

Test Accounts (README.md:66)

Admin — admin@cqi.test / Admin@123456 • Manager — manager@cqi.test / Manager@123456 • User — user@cqi.test / User@123456

Roles restored on sign-in via sync_demo_role() — survives supabase-schema.sql re-run.